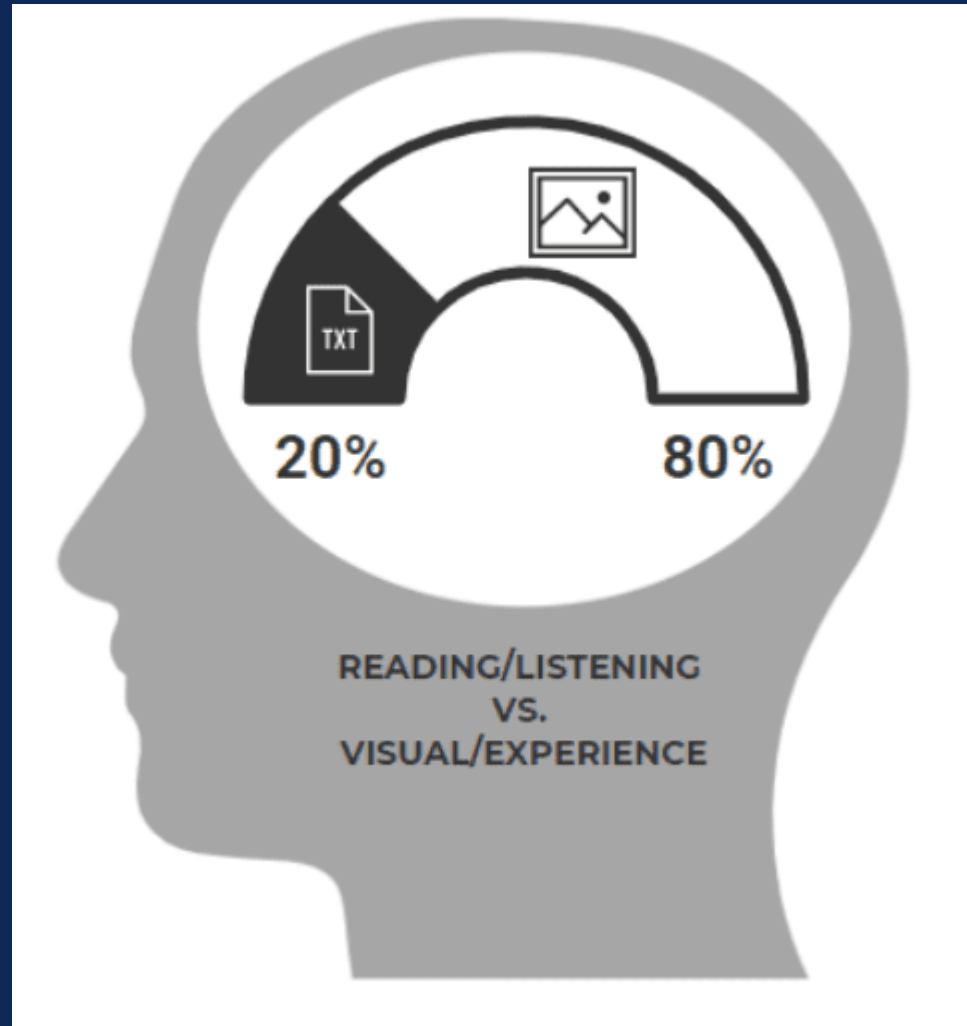


DATA VISUALIZATION: KEY TO EFFECTIVE COMMUNICATION

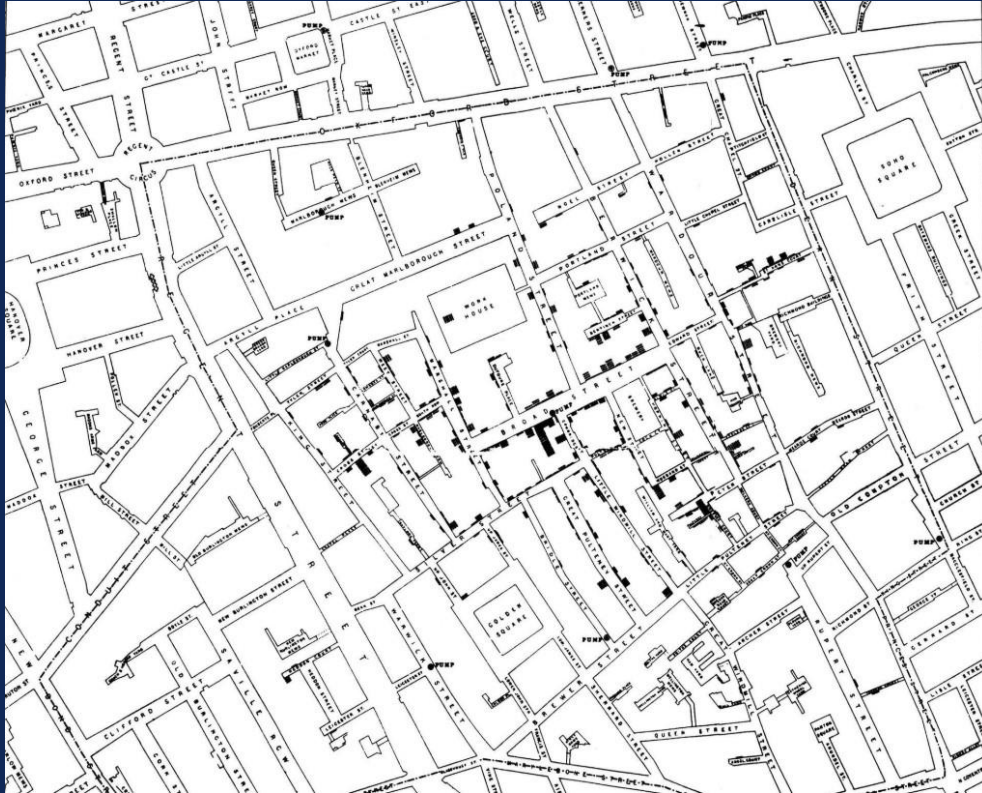
Garrett Bullock, PT, DPT, DPhil

Picture is Worth a 1000 Words

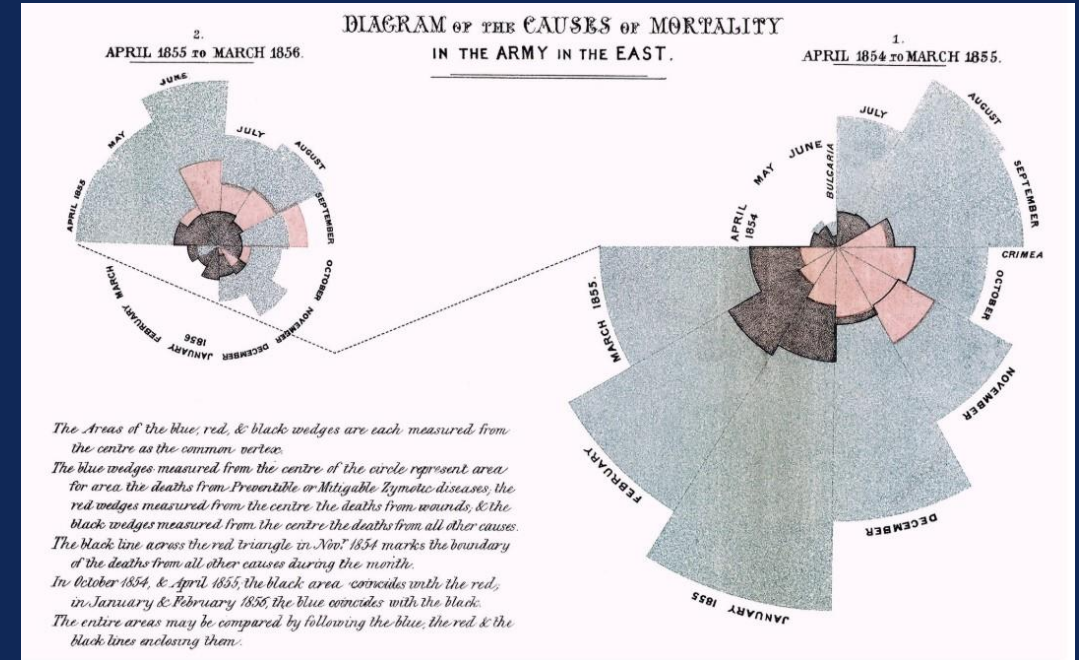


If data is king, data visualization is the kingmaker

DATA VISUALIZATION: A HISTORY



Jon Snow
Cholera Epidemic of 1854

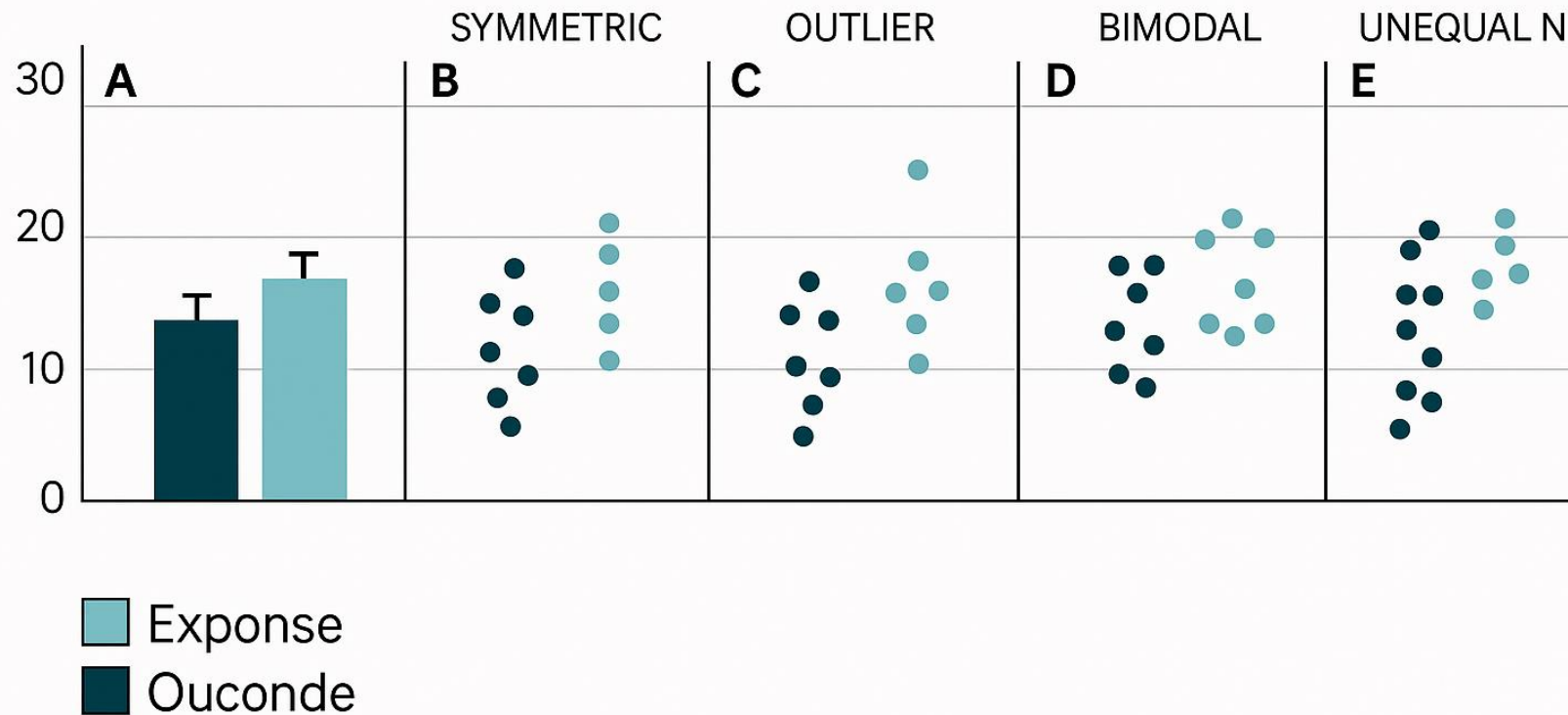


Florence Nightingale
Crimea War 1855

Effective Scientific Communication Starts with Visuals

Hidden in the bars

Data revealed in scatterplots may be masked within a bar chart.



SOURCE: T.J. VERHOEF ET AL / PLOS BIOLOGY 2016

SW INFOGRAPHIC / KNOWABLE

Even Brooklyn 99 Understands Scientific Visualizations are Key!

Rosa: Relax. It's not a big deal. You were just flirting.

Captain Holt: No, if I were flirting, I would have sent them a scatter plot of educational attainment versus caloric intake in Jacobin France. This is as if I've sent them a bar graph.

Rosa: Are bar graphs bad?

Captain Holt: It reduces robust data sets to a single point. Get your head out of your ###!



How Should We Think About Scientific Visualization?

- Clear & related to the information you want to present
- Should be able to skim (at a glance) the information & understand the overarching message

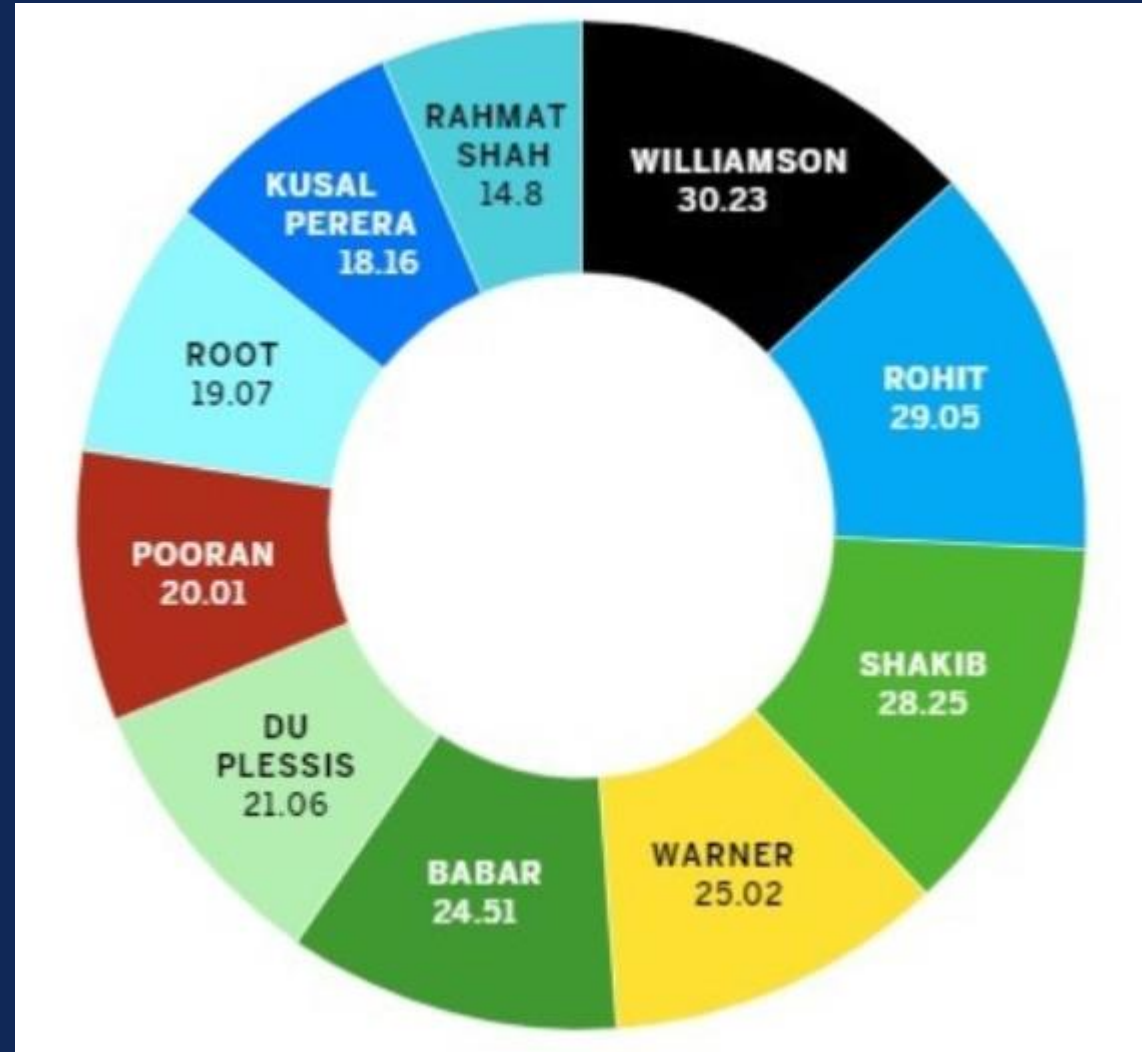
How Should We Think About Scientific Visualization?

- Should be intuitive – If you have to ask a lot of questions for orientation, it is not effective scientific communication
- Implicit hierarchy of relationships (e.g., sport position, different hospitals, etc.)
- Should not hide or obscure the data itself

Some Rules to Avoid Bad Scientific Visualization

- Know the audience
- Simpler is better
- Explicitly label your data
- Don't reduce data points to one point
- Don't add too much color
- Comparisons should be side by side
- Axes should be exact between comparisons

Some Examples of Bad Visualizations



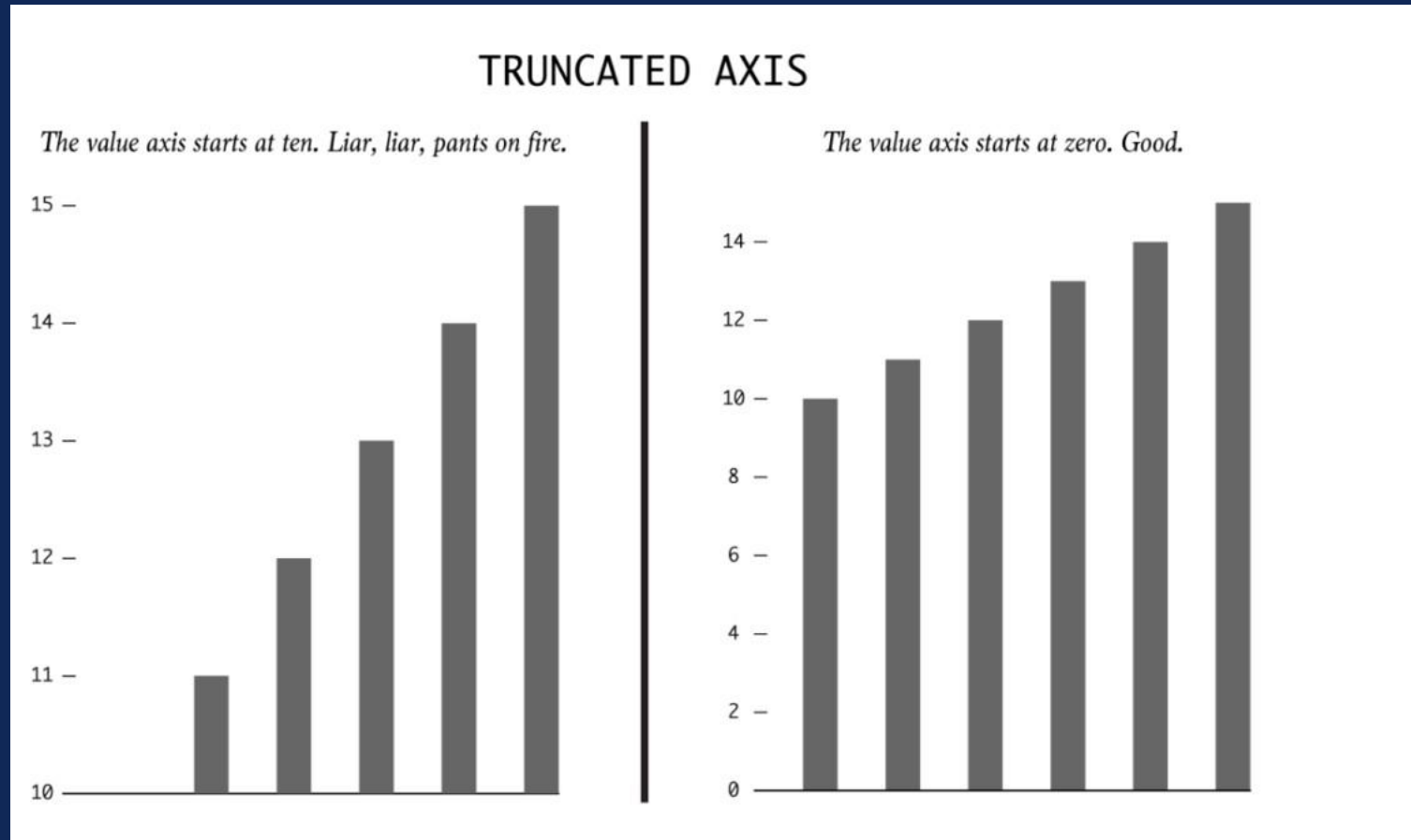
Some Examples of Bad Visualizations



SOME EXAMPLES OF BAD VISUALIZATIONS



SOME EXAMPLES OF BAD VISUALIZATIONS



SOME EXAMPLES OF BAD VISUALIZATIONS



SOME EXAMPLES OF BAD VISUALIZATIONS

ODD CHOICE OF BINNING

*Two bins. What's really in the 1+ category?
Might be hiding something.*



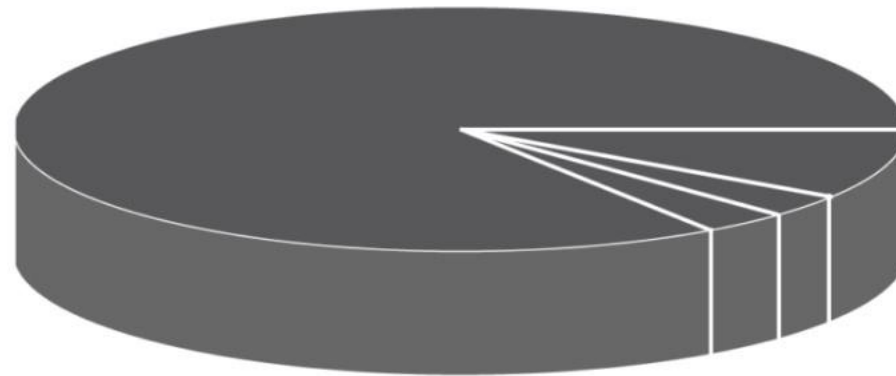
That's better. It can show more variation.



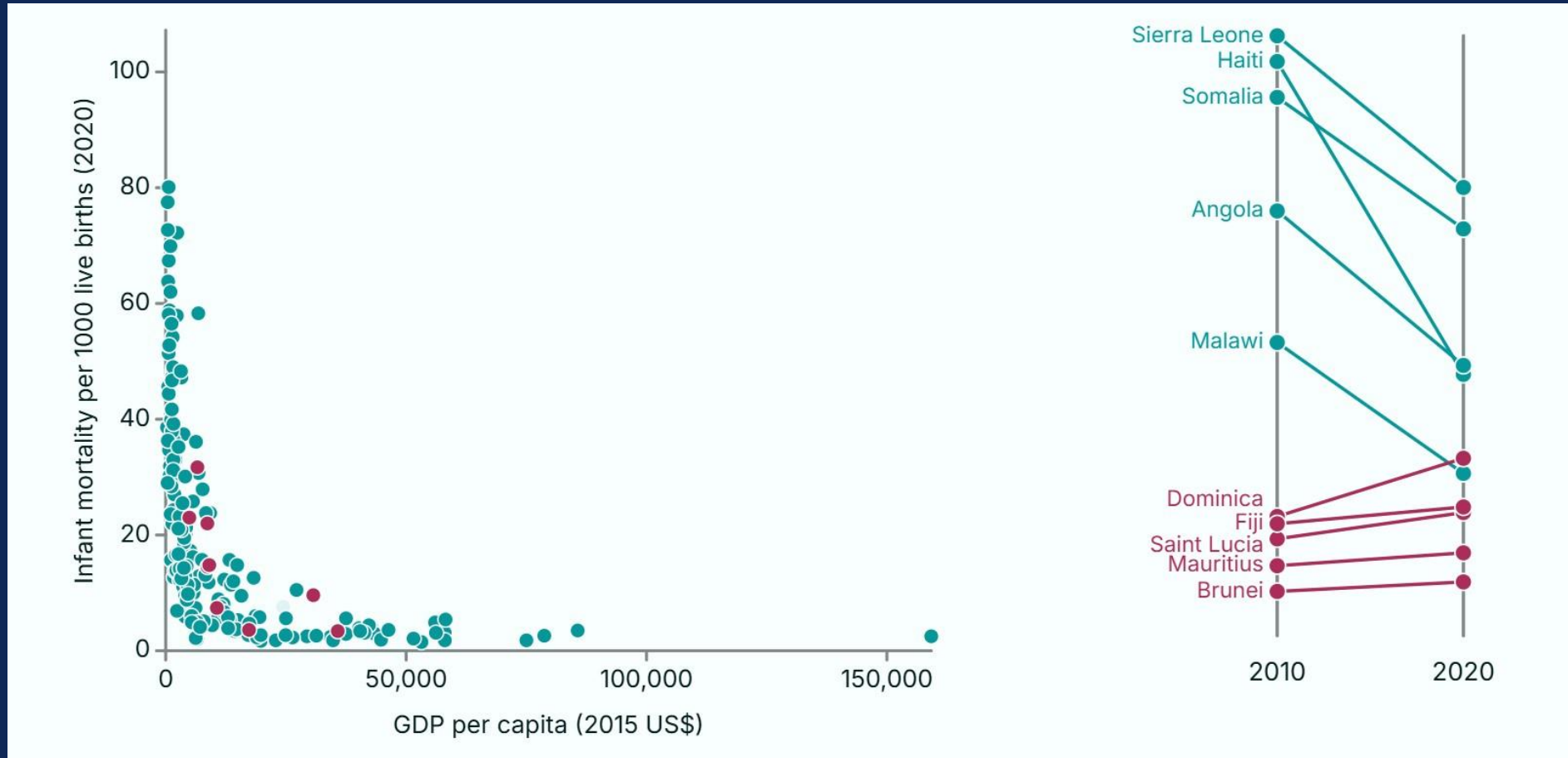
SOME EXAMPLES OF BAD VISUALIZATIONS

EXTRA DIMENSION JUST BECAUSE

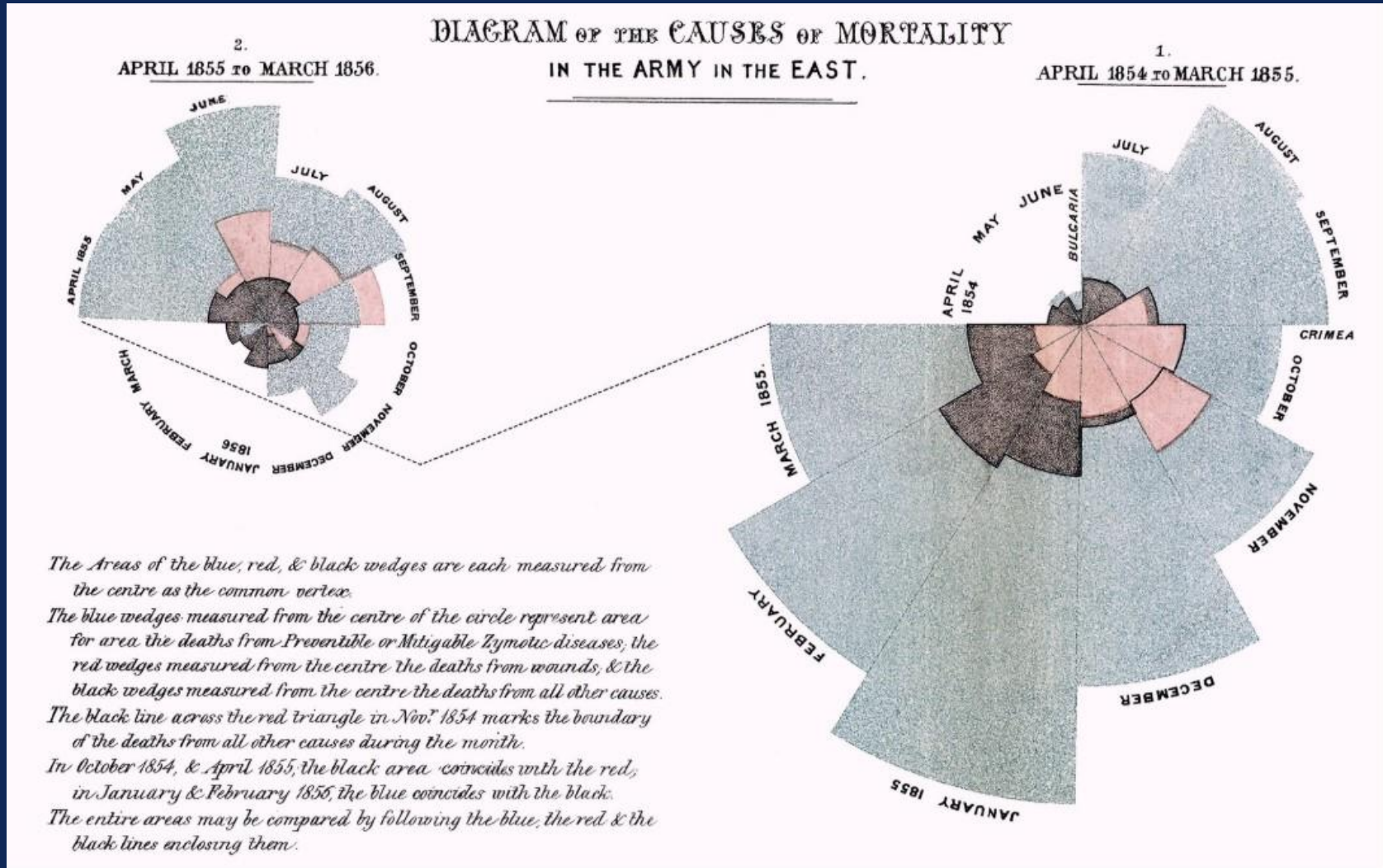
Just say no.



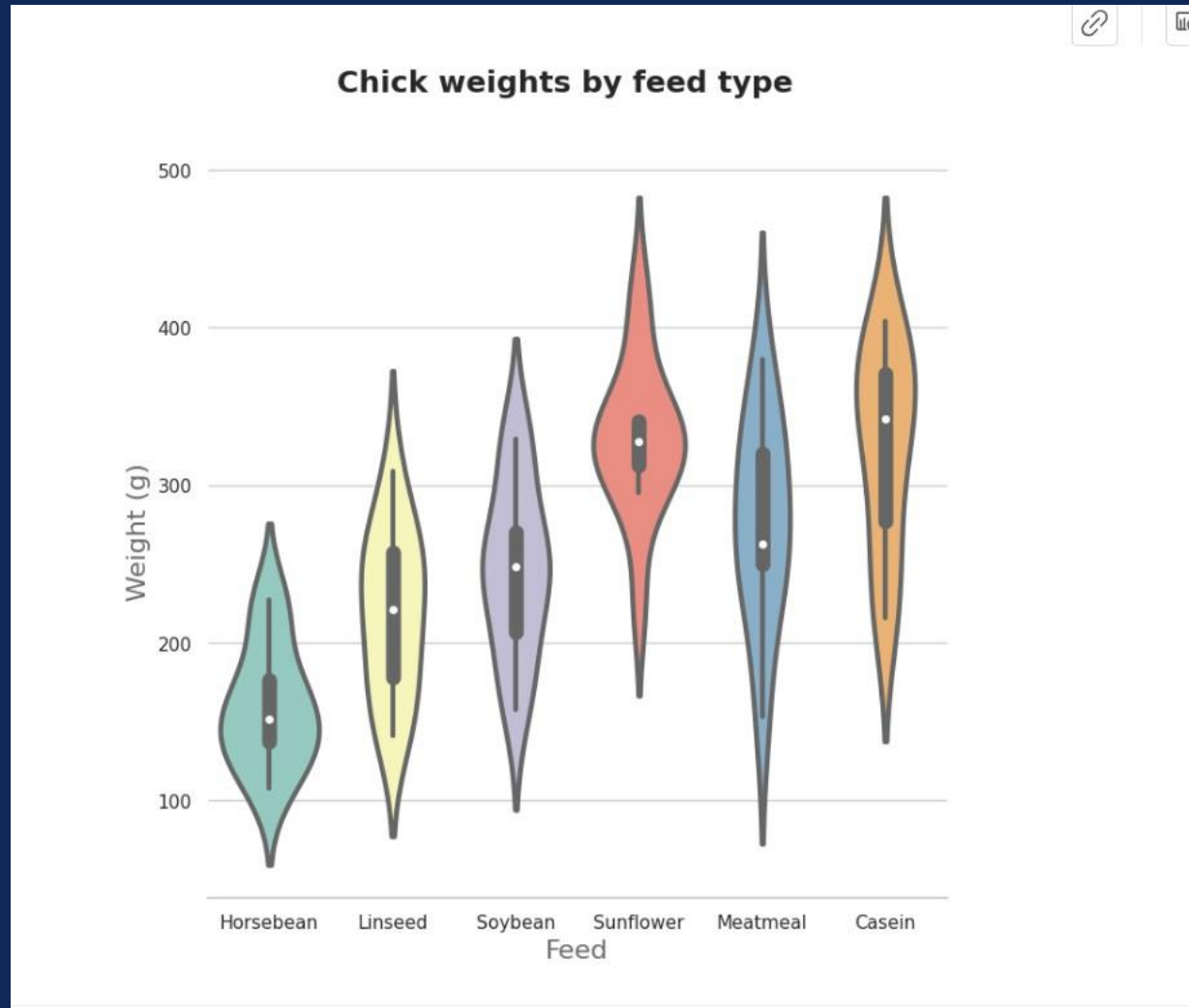
SOME EXAMPLES OF GOOD VISUALIZATIONS



SOME EXAMPLES OF GOOD VISUALIZATIONS



SOME EXAMPLES OF GOOD VISUALIZATIONS



UNDERSTANDING THE BASICS OF DATA VISUALIZATION CODING

- ggplot2 library is the 'foundation'
- The major 'pieces' of creating a base visualization
 - Dataframe
 - X & Y
 - Subgroups
 - How the data will be displayed

UNDERSTANDING THE BASICS OF DATA VISUALIZATION CODING

- Dataframe
 - `ggplot(dataframe, aes())`
- X & Y
 - `ggplot(dataframe, aes(x, y))`
- Subgroups
 - `ggplot(dataframe, aes(x, y, color = z))`
- How the data will be displayed
 - `ggplot(dataframe, aes(x, y, color = z)) + geom_point()`

UNDERSTANDING THE BASICS OF DATA VISUALIZATION CODING

'Base' Code

- `ggplot(dataframe, aes(x, y, color = z))) + geom_point()`

Add Color to your 'points

- `ggplot(dataframe, aes(x, y, color = z))) +
 geom_point(color = "blue")`

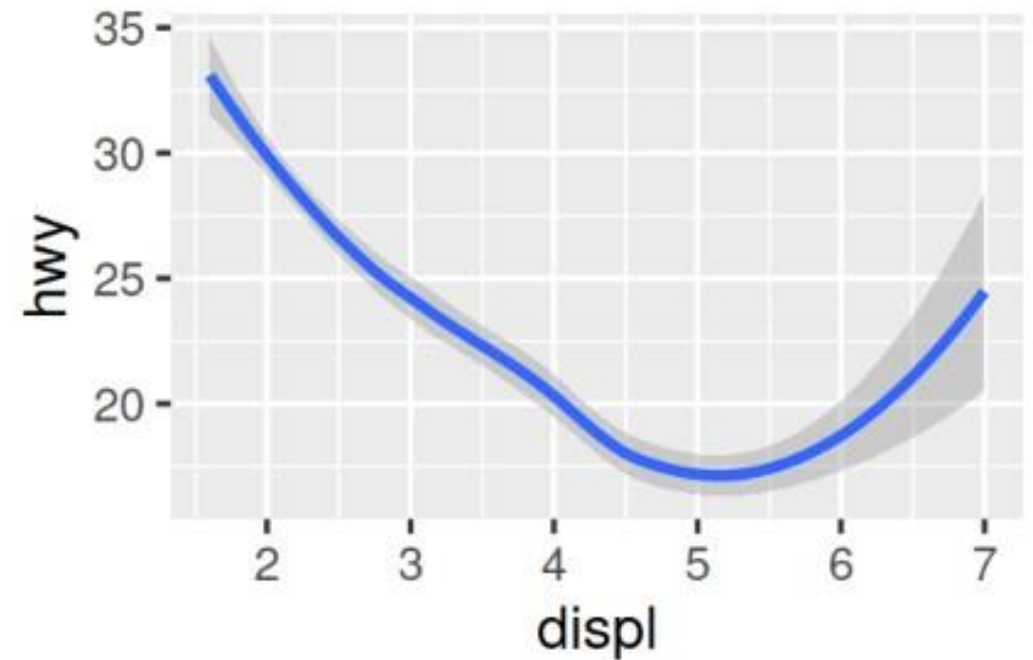
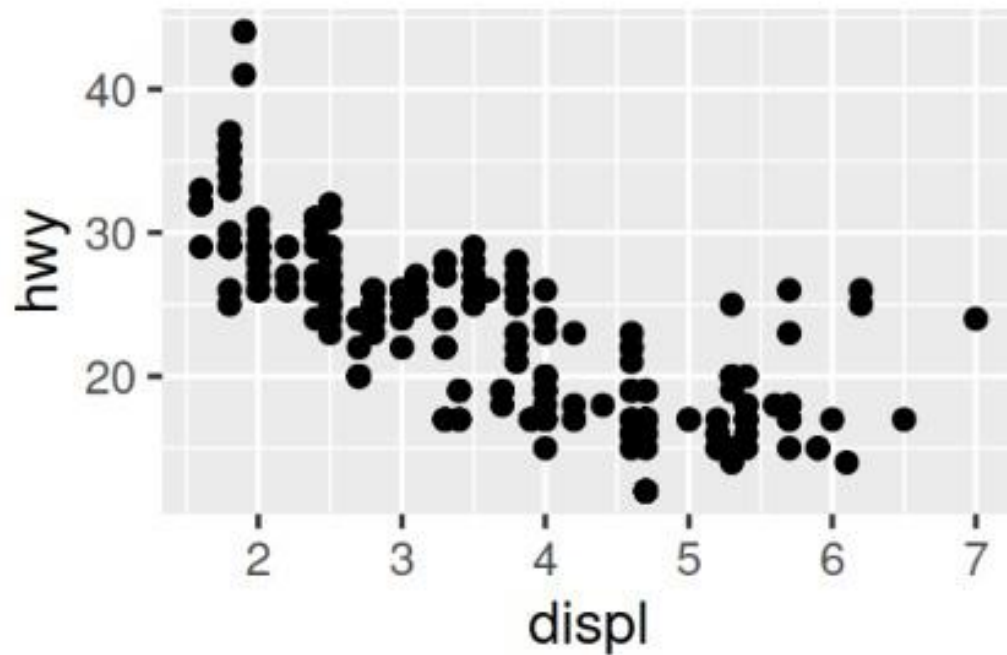
Changing Size or shape to you 'oints

- `ggplot(dataframe, aes(x, y, color = z))) +
 geom_point(color = "blue", shape = 1, size = 1)`

POINTS VS. 'SMOOTH' REGRESSION

```
ggplot(mpg, aes(x = displ, y = hwy)) + geom_point()
```

```
ggplot(mpg, aes(x = displ, y = hwy)) + geom_smooth()
```



ADDING TITLES TO AXES

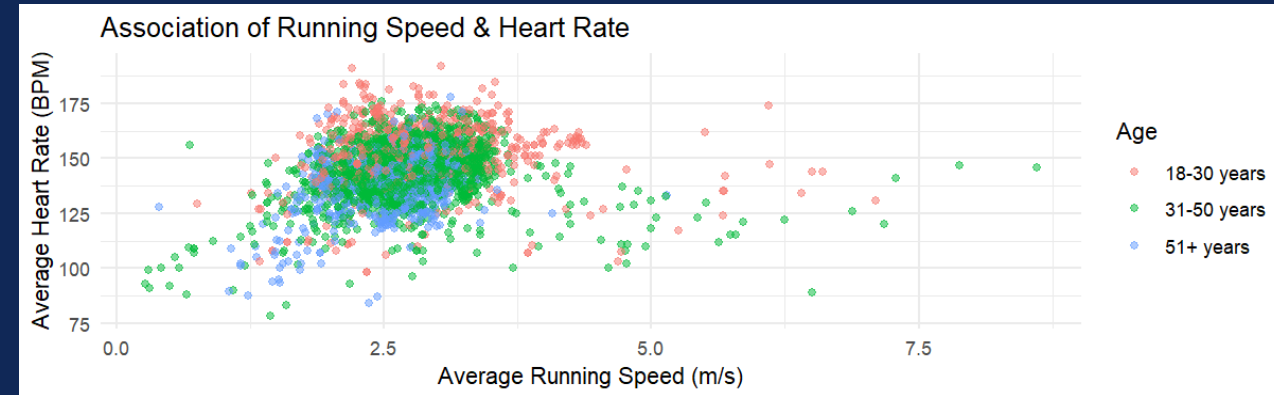
```
ggplot(dataframe, aes(x, y, color = z)) +  
  geom_point(alpha = 0.5) + # 50% opaque  
  xlab('Average Running Speed (m/s)') +  
  ylab('Average Heart Rate (BPM)') +  
  ggtitle("Association of Running Speed & Heart Rate")
```

ADDING SPECIFIC LEGEND

```
ggplot(dataframe, aes(x, y, color = z)) +  
  geom_point(alpha = 0.5) + # 50%  
  opaque
```

```
  xlab('Average Running Speed (m/s)') +  
  ylab('Average Heart Rate (BPM)') +  
  ggtitle("Association of Running Speed  
& Heart Rate") +
```

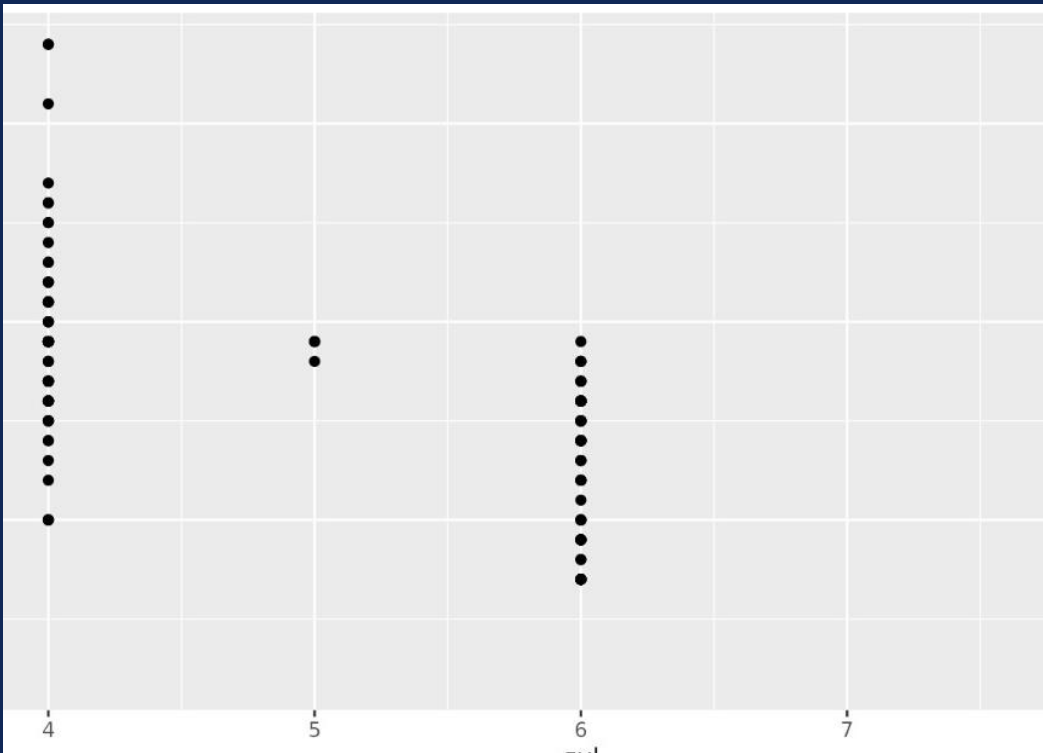
```
  labs(color = "Age") +  
  scale_color_discrete(labels = c("18-30  
years", "31-50 years", "51+ years")) +
```



JITTER!

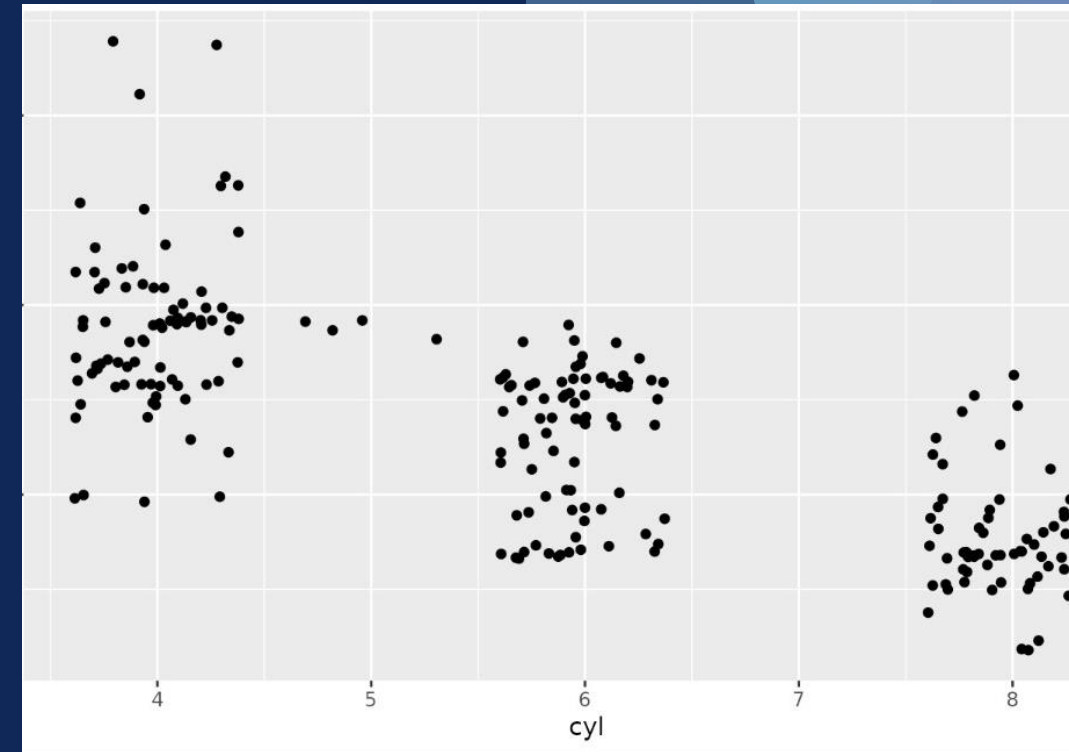
‘Regular Point’

```
ggplot(mpg, aes(cyl, hwy)) +  
geom_point()
```



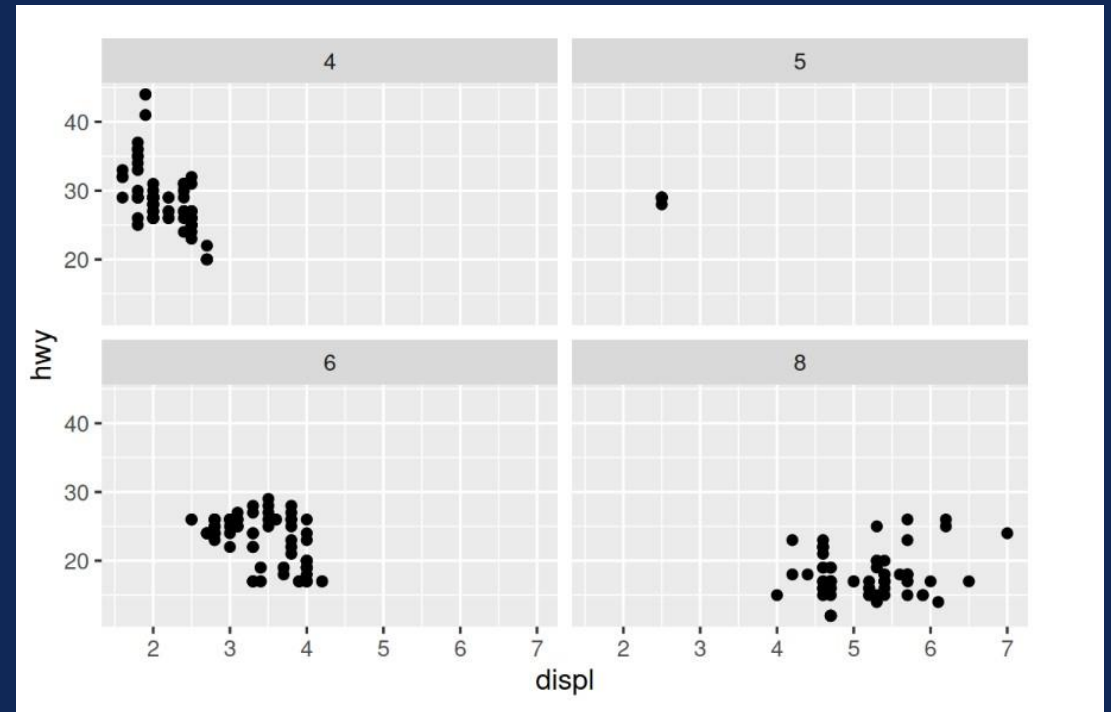
‘Jitter’

```
ggplot(mpg, aes(cyl, hwy)) +  
geom_jitter()
```



ADDING SIDE-BY-SIDE COMPARISONS

○ `ggplot(mpg, aes(x = displ, y = hwy)) + geom_point() + facet_wrap(~cyl)`



Final Thoughts

- Scientific communication starts with visualization
- Be clear
- Don't hide data
- Use color to expand the story, not for glitz